except for the grade, thus minimizing storage. Storage needed for each student is about nine words.

Listing 11.1 DCC.sol

```
contract DICCertification{
    uint constant private MINIMUM_GPA_REQUIRED = 250;

    struct Student {                                                    ❶
        uint personNumber;

        uint prereq115;
        uint prereq116;

        uint core250;
        uint core486;
        uint core487;

        uint domainSpecificCourse;
        uint domainSpecificGrade;
        uint capstoneCourse;
        uint capstoneGrade;

    }

    mapping(address => Student) registeredStudents;                     ❷

    event preRequisiteSatisfied(uint personNumber);                     ❸
    event coreCoursesSatisfied(uint personNumber);                      ❸
    event GPARequirementSatisfied(uint personNumber);                   ❸
    event projectRequirementSatisfied(uint personNumber);               ❸
    event domainRequirementSatisfied(uint personNumber);                ❸
    event GPA(uint result);                                             ❸

//-----------------------------------------
// Modifiers
//-----------------------------------------
    modifier checkStudent(uint personNumber) {                          ❹
        require(registeredStudents[msg.sender].personNumber == 0,
                                  "Student has already registered");
        _;}

    modifier validStudent(){  //#D
        require(registeredStudents[msg.sender].personNumber > 0,
                                     "Invalid student");
        _;}

//-----------------------------------------
// Functions
//-----------------------------------------
    function registerStudent(uint personNumber) public
                                 checkStudent(personNumber) {           ❺
```